

Enhancing carbon sequestration through flocculation of harmful algal blooms by modified clay technology

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ABSTRACT

Global efforts to mitigate climate change emphasize the critical need to enhance carbon sinks. Harmful algal blooms (HABs) areas represent hot spots for labile organic carbon production, yet their carbon sequestration capacity is diminished by rapid microbial mineralization. Modified clay (MC) technology, used for HABs mitigation, could enhance the sedimentation and influence the transformation dynamics of alga-derived organic carbon. Nevertheless, the impacts of MC on the mineralization of alga-derived organic carbon and the mechanisms involved remain inadequately understood. In this study, employing ¹³C-labeled batch incubation experiments, we demonstrated that alga-derived organic carbon undergoes rapid mineralization, with 46%–59% of the total organic carbon (TOC) being mineralized within 30 days. MC substantially reduced the mineralization rate of alga-derived organic carbon from 0.68–0.80 mg/(L·d) to 0.09 mg/(L·d), thereby boosting organic carbon sequestration potential by approximately 55–70% compared to the control group. Multiple lines of evidence, including microscopic imaging, fluorescence spectroscopy, and microbial analysis, revealed that MC promoted organic carbon downward export, reduced bioavailability through flocculation and encapsulation, and suppressed heterotrophic bacteria. Notably, the MC treatment group exhibited a significant reduction, with the abundance of heterotrophic bacteria decreasing by approximately 60% and the functional genes associated with microbial mineralization dropping by 50%. Overall, this study presents direct evidence and mechanistic insights that demonstrate the feasibility of employing MC to enhance carbon sequestration in mitigating HABs.

1. Introduction

Enhancing oceanic carbon sequestration is increasingly recognized as a viable strategy for mitigating global climate change. The oceans represent the largest active carbon reservoir, holding 93% of the Earth's total carbon reserves, a quantity 20 to 50 times greater than that of terrestrial and atmospheric carbon reservoirs, respectively (Friedlingstein et al., 2020). Marine algae play a pivotal role in the global carbon cycle, with recent studies highlighting their substantial potential for carbon sequestration (Sarwer et al., 2022; Tréguer et al., 2018; Zhang et al., 2023). It is estimated that marine algae fix over 10 billion tons of carbon annually, accounting for approximately 50% of global primary production (Lewis et al., 2020; Krause-Jensen and Duarte, 2016; Sarmento and Gasol, 2012). However, the rapid proliferation and

aggregation of microalgae often trigger large-scale harmful algal blooms (HABs), which are occurring with increasing frequency worldwide (Dai et al., 2023; Gobler, 2020). Notably, following the dissipation of HABs, a considerable quantity of alga-derived organic carbon is rapidly degraded and mineralized by heterotrophic bacteria (Bittar et al., 2015), markedly diminishing carbon sequestration capability (Wang et al., 2021) while inducing hypoxia and severely disrupting marine ecosystem (Wang et al., 2017).

HABs areas represent hot spots for labile organic matter production, thereby playing a significant role in oceanic carbon cycling (Shen et al., 2016). Typically, blooming microalgae contribute to carbon sequestration primarily through the biological pump (BP) and microbial carbon pump (MCP) (Vidal-Melgosa et al., 2021; Zhang et al., 2018). However, alga-derived organic carbon undergoes rapid mineralization in the

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water column, with only a small fraction ultimately sequestered (Armbrust, 2009; Tréguer et al., 2018, 2021). It is estimated that a mere 0.1–0.3% of algae-derived organic carbon from the euphotic zone reaches the seafloor (Marsay et al., 2015). The release and subsequent mineralization of organic carbon during its descent are influenced by a multitude of various chemical, physical, and biological factors, with aggregation and sinking rates being pivotal determinants of organic carbon export to the deep ocean (Yuan and Liu, 2021). To tap the sequestration potential of algal-derived organic carbon, diverse strategies have been proposed to regulate BP and MCP processes. Notably, the concept of mineral-enhanced carbon pump has emerged as a widely acknowledged conceptual framework for enhancing carbon sequestration (Xiao et al., 2025). This approach boosts BP efficiency by accelerating the transport of organic carbon to the seabed (Yuan and Liu, 2021; Zhang et al., 2023), while reducing the bioavailability of algae-derived organic carbon through interactions with clay minerals (Landa et al., 2016; Zhang et al., 2023).

Modified clay (MC) technology is widely acknowledged as an effective approach for mitigating HABs through flocculation. Currently, it has been extensively applied, particularly in China's coastal waters, owing to its cost-effectiveness, high efficiency, and environmental compatibility (Song et al., 2021; Yu et al., 2017). During the HABs mitigation with MC, a substantial fraction of algal-derived organic carbon is rapidly transported to deeper waters, prompting significant scientific interest in its ultimate fate. Notably, MC may confer unique advantages among mineral-enhanced carbon pump strategies. It can rapidly mitigate HABs through flocculation of blooming microalgae, forming 'alga-MC flocs' through charge neutralization and net capture, which markedly amplifies vertical organic carbon export (Yu et al., 2017). Comprising kaolinite clay and poly-aluminum chloride (PAC), MC possesses robust adsorptive properties that profoundly modulate the formation of mineral-associated organic carbon (Rakhsh et al., 2020; Sokol et al., 2019). This fraction constitutes a more recalcitrant pool of organic carbon (Lavalée et al., 2020), with the potential to persist for decades to centuries within marine sediments (Blattmann et al., 2019; Rakhsh et al., 2020). Additionally, accumulating evidence underscores aluminum's role in promoting organic carbon stabilization via dissolution inhibition and enhanced aggregation (Yuan and Liu, 2021; Zhou et al., 2021). Research indicates that MC demonstrates substantial efficacy in removing a variety of HAB organisms and algae-derived organic carbon, emphasizing its potential for applications in large-scale carbon sequestration (Anderson et al., 2025; Yu et al., 2017). Nevertheless, the fate of algal-derived organic carbon followed flocculation and sedimentation facilitated by MC remains insufficiently understood. To date, research focusing on the effects of MC on the mineralization of algal-derived organic carbon is notably scarce, and the underlying mechanisms by which MC influences this process remain largely unexplored.

To this end, using isotopically labelled incubation experiments, this study aims to (i) quantitatively determine the mineralization rate of algal-derived organic carbon and assess the inhibitory effects of MC on carbon mineralization; and (ii) explore the mechanism by which MC enhances carbon sequestration, focusing on physical protection and microbial regulation. Overall, this study highlights the previously overlooked yet critical role of MC, offering valuable insights into the viability of a mineral-enhanced carbon sequestration strategy.

2. Materials and methods

2.1. Experimental materials

Skeletonema costatum and *Prorocentrum donghaiense* were cultured at the Institute of Oceanology, Chinese Academy of Sciences. To avoid external organic carbon interference, *S. costatum* and *P. donghaiense* were cultivated in L1 medium prepared with artificial seawater (Table S1) (Berges et al., 2001) under a light intensity of 60 μmol

photons $\text{m}^{-2} \text{s}^{-2}$ and a 12 h light/12 h dark photoperiod at $20 (\pm 1) ^\circ\text{C}$. The polyaluminium chloride-MC (PAC-MC, MCI, detailed chemical composition in Table S2) was made by mixing kaolin and polyaluminum chloride in a 5:1 ratio, according to the method described by Yu et al. (1994). Prior to the incubation experiment, a 200g/L MCI stock solution was prepared in Milli-Q ultrapure water.

2.2. Batch incubation experiments

The mineralization rates of algal-derived organic carbon were assessed by using the isotope dilution method (Dong et al., 2017, 2019). In this study, the total production of dissolved inorganic carbon (DIC, including CO_2 , HCO_3^- and CO_3^{2-}) as indicative of mineralization. During the incubation experiment, $\text{NaH}^{13}\text{CO}_3$ ($\geq 99\%$, Sigma-Aldrich Co.) was introduced into *S. costatum* (1.1×10^8 cells/L) and *P. donghaiense* (1.0×10^8 cells/L) cultures, both of which were in the exponential growth phase. This ensured that the labeled ^{13}C comprised 10–15% of the DIC concentration in the incubation system, with an actual calculation of 10%. Afterward, these algal cultures were then distributed into 90 cell culture flasks (200 ml each) for dark cultivation to mimic the decline of HABs and prevent the photodegradation of organic carbon from interfering with the experiment. Throughout the cultivation process, the incubation system was maintained in a closed environment, preventing any gas exchange with the external atmosphere.

The long-term mineralization of the algal cultures without the addition of MC was considered to establish the control group. In the MC treatment group, MCI suspension (200 g/L) was added to the flasks, achieving a final MCI concentration of 0.5 g/L. Sampling was conducted at intervals of 0 h, 1 h, 3 h, 6 h, 9 h, 12 h, 24 h, 3 d, 5 d, 7 d, 10 d, 15 d, 20 d, 25 d, and 30 d. At each interval, three samples were collected from independent replicates of each experimental treatment to ensure robustness and reliability. Samples were taken and analyzed for total organic carbon (TOC), dissolved organic carbon (DOC), DIC, $\delta^{13}\text{C}$ -DIC, algal density, nutrients, pH, dissolved oxygen (DO), and microbial abundance and community. Furthermore, the excitation–emission matrix was used to trace the changes in the molecular composition and biodiversity of algal-derived organic matter.

2.3. Analytical methods

Concentrations TOC and DIC were measured using a TOC analyzer (Multi N/C 2100S, Analytik Jena, Germany), with TOC determined by the high-temperature catalytic combustion method. The $\delta^{13}\text{C}$ -DIC values were analyzed using an isotope ratio mass spectrometer (MAT252, Thermo Scientific, USA). Nutrients, including NO_3^- , NO_2^- , NH_4^+ , PO_4^{3-} , and SiO_3^{2-} , were analyzed with a SKALAR nutrient autoanalyzer through spectrophotometry method (San⁺⁺ Skalar Ltd., Netherlands). pH and DO were measured with a pH meter (Mettler Toledo, Seven Excellence) and a DO meter (YSL, Pro Solo), respectively. The density of algal cell was measured by cell counting under an inverted microscope (IX71, Olympus, Japan).

2.4. Microbial abundance, community composition and function

To elucidate the microbial mechanisms by which MC influences the mineralization of algae-derived organic carbon, we quantified three critical microbial attributes: biomass, community structure, and functional potential. Microbial abundance was determined by staining samples with SYBR Green I (final dilution 1:10,000) for 15 min in the dark, followed by enumeration using an inverted fluorescence microscope (OLYMPUS IX71, Japan). Microbial community composition was assessed via 16S rRNA gene amplicon sequencing (detailed protocols in Text S1). Differential abundance analysis was conducted using LEfSe, with linear discriminant analysis (LDA) thresholds set at LDA score > 4 to identify biomarker taxa distinguishing treatment groups. Functional profiles of the microbial communities were predicted by mapping

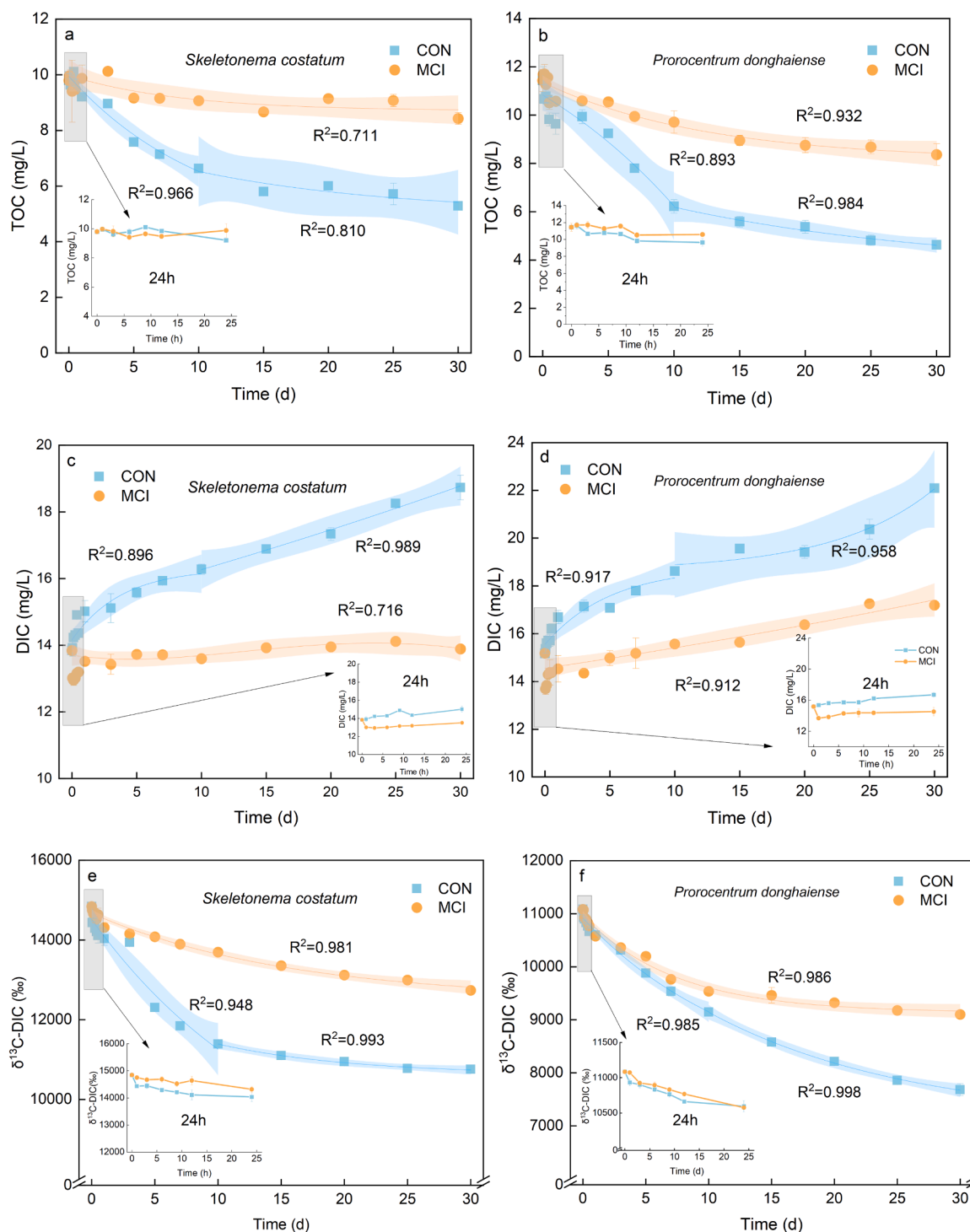


Fig. 1. Variations in the TOC, DIC, and $\delta^{13}\text{C}$ -DIC during the batch incubation period (a, c, and e: *S. costatum*; b, d, and f: *P. donghaiense*).

against the FAPROTAX database (Functional Annotation of Prokaryotic Taxa).

2.5. Characterization of DOM

Excitation–emission matrix (EEM) of the fluorescent dissolved organic matter (FDOM) was measured using a fluorescence spectrophotometer (F-4600, Hitachi), scanning excitation wavelengths (Ex) from 200 to 600 nm and emission wavelengths (Em) from 220 to 550 nm at 5 nm intervals. Milli-Q ultrapure water was employed as the blank.

Subsequent data processing, encompassing blank subtraction, Raman normalization, UV correction, Raman–Rayleigh scattering removal, and parallel factor analysis (PARAFAC), was performed with the drEEM toolbox (version 0.2.0) in MATLAB R2024a.

2.6. Electron microscopy of algal cell–MC flocs

Following the addition of MC, the settled flocs were transferred to a 50 mL centrifuge tube, and allowed to stabilize for 15 minutes. Subsequently, they were washed three times with phosphate-buffered saline

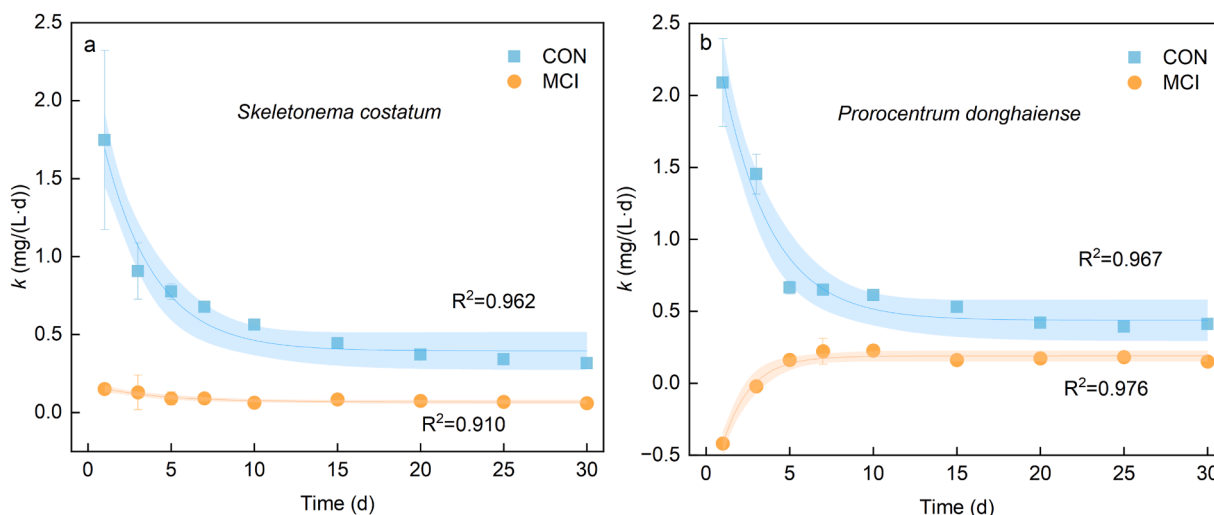


Fig. 2. Mineralization rates of alga-derived organic carbon during the incubation period.

(PBS) to remove any interference from excess NaCl crystals. The floc samples were then snap-frozen in liquid nitrogen for 15 min and lyophilized at -50°C for 24 h to minimize structural distortion. The resulting dried samples were stored under desiccated conditions prior to preparation for imaging. The samples were mounted on conductive adhesive, sputter-coated with gold, and examined via a JSM-IT800 scanning electron microscope (JEOL, Japan) operated at 5 kV and $15,000\times$ magnification to characterize the architecture and binding mode of the algal cell-MC flocs.

2.7. Statistical analysis

The $\delta^{13}\text{C}$ values were calculated according to the Pee Dee Belemnite (PDB) standard. In this study, organic carbon mineralization followed the decay equation described by (Dai et al., 2009). We integrated measurements of DIC regeneration with temporal variations in $\delta^{13}\text{C}$ signatures to precisely quantify the mineralization kinetics of algal-derived organic carbon.

$$k = \frac{C_0 - C_t}{t} \times \frac{\log(H_0 C_t / H_t C_0)}{\log(C_0 / C_t)} \quad (2.1)$$

Where C_0 denotes the initial DIC concentration (mg/L), C_t is the DIC concentration at cultivation time t (mg/L), H_0 is the initial $^{13}\text{C}_{\text{DIC}}$ concentration (%), H_t is the $^{13}\text{C}_{\text{DIC}}$ concentration (%) at cultivation time t , and t is the incubation time.

Sequencing data were analyzed via principal component analysis (PCA) with the ade4 and ggplot2 packages in R. Complementarily, redundancy analysis (RDA) of environmental variables was performed with the vegan package. All samples in this study were processed in three parallel replicates, with error bars calculated accordingly. Statistical significance was established using a threshold of $p < 0.05$. Functional gene copy numbers were quantified on the QIIME cloud platform, incorporating USEARCH for filtering low-quality sequences and optimized merging of paired-end reads. Representative operational taxonomic unit (OTU) sequences were taxonomically classified at a 97% similarity cutoff. Visualizations were generated using Origin 2025, Adobe Illustrator 2024, MATLAB R2024a, and Chiplot.

3. Results

3.1. Variations in OC, DIC, and $\delta^{13}\text{C}_{\text{DIC}}$ during incubation

During the incubation experiments, both the control and MC treatment groups maintained a fundamental carbon mass balance (Fig. S1).

Notably, TOC concentrations declined progressively over time in both groups. For *S. costatum*, the TOC concentration in control group decreased from 9.80 to 5.80 mg/L after 15 days and further to 5.29 mg/L by 30 days, corresponding to 46% mineralization. In contrast, the MCI treatment exhibited a slower decline, with TOC concentrations decreasing from 9.80 to 8.43 mg/L by 30 days, resulting in 14% mineralization (Fig. 1a). Similarly, for *P. donghaiense*, TOC concentration in control group decreased from 11.43 to 5.58 mg/L after 15 days, and further to 4.63 mg/L by 30 days, corresponding to 59% mineralization. In the MC treatment group, the TOC concentration gradually decreased from 11.43 to 8.95 mg/L after 30 days, yielding a mineralization ratio of 27% (Fig. 1b).

The concentrations of DIC in the control group of *S. costatum* increased rapidly from 13.84 to 16.88 mg/L within 15 days, followed by a more gradual rise to 18.73 mg/L. By comparison, the MCI treatment group exhibited a negligible increase, from 13.84 to 13.89 mg/L by day 30 (Fig. 1c). For *P. donghaiense*, DIC concentrations in the control group demonstrated a similar pattern, with a sharp increase from 15.18 to 19.56 mg/L within 15 days, and a subsequent slower increase to 22.09 mg/L by day 30. In contrast, the MC treatment group initially showed a decrease to 13.69 mg/L immediately after the addition of MC, followed by a gradual increase to 17.19 mg/L by day 30 (Fig. 1d). Similar trends were observed in the $\delta^{13}\text{C}_{\text{DIC}}$ values. For *S. costatum*, the control group exhibited a significant decline from 14,840‰ to 12,306‰ over the initial 15 days, continuing to 10,756‰ by day 30, whereas the MC treatment group experienced a much less pronounced decrease, from 14,840‰ to 12,736‰ over the entire 30 days (Fig. 1e). A comparable trend was observed in the *P. donghaiense* experiment, where the control $\delta^{13}\text{C}_{\text{DIC}}$ values decreased from 11,084‰ to 7,682‰ by day 30, while the MC treatment group experienced a more moderate reduction to 9,100‰ over the incubation period (Fig. 1f).

3.2. Mineralization rate of algal-derived organic carbon

In this study, the mineralization rate of algal-derived organic carbon was quantified by integrating DIC regeneration with variation in $\delta^{13}\text{C}_{\text{DIC}}$ (Formula 2.1). During the batch incubation experiments, the mineralization rates in the control groups of *S. costatum* (Fig. 2a) and *P. donghaiense* (Fig. 2b) ranged from 0.32 to 1.75 mg/(L·d) and 0.41 to 2.09 mg/(L·d), respectively. These rates exhibited a gradual decline as the incubation period progressed. The average mineralization rates were 0.68 mg/(L·d) for *S. costatum* and 0.80 mg/(L·d) for *P. donghaiense*, respectively (Fig. S5). Notably, the MC treatment groups exhibited a significant reduction in the mineralization rate of algal-derived organic

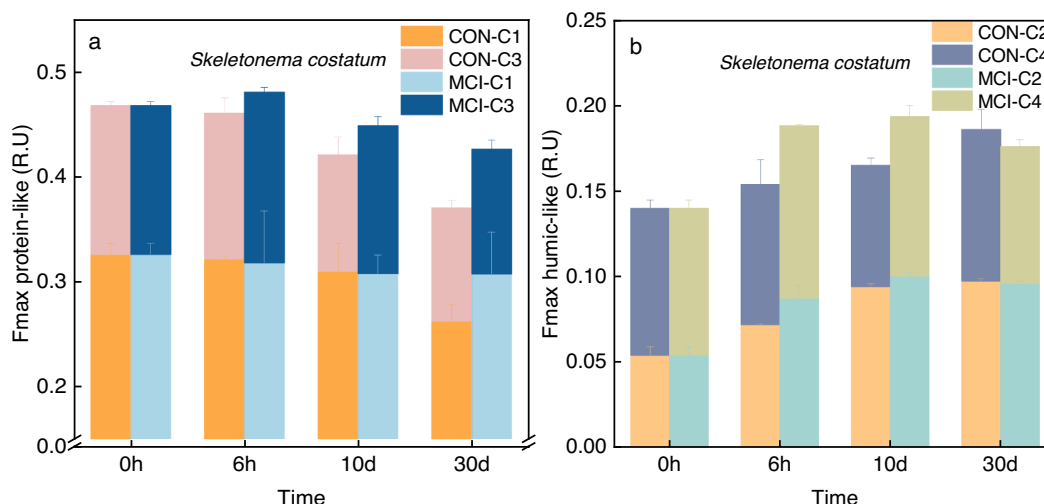


Fig. 3. DOM fluorescence components and fluorescence intensity of *S. costatum* (a: protein-like fluorescence intensity, b: humic-like fluorescence intensity).

carbon, with average mineralization rates of 0.09 mg/(L·d) for both *S. costatum* and *P. donghaiense* (Fig. 2).

3.3. Characteristics of the active fluorescent components

The PARAFAC analysis identified four distinct fluorescent components, each defined by its characteristic maximum excitation and emission wavelengths (Fig. S3). Components C1 (280 nm/310 nm) and C3 (320 nm/345 nm) are typically identified as tyrosine-like and tryptophan-like fluorophores, respectively, both are classified as protein-like substances that reflect the relative abundance of bioavailable DOM. In contrast, components C2 (240 nm/510 nm) and C4 (370 nm/390 nm) correspond to fulvic-like and humic-like acids, respectively, which are used as proxies for the relative concentrations of refractory DOM. Notably, MC-induced flocculation significantly influenced the dynamics of these fluorescent DOM components. In the control group, the intensity of protein-like fluorescence consistently decreased from 0.47 to 0.37 R.U. throughout the incubation period. In contrast, the protein-like fluorescence in the MC treatment groups remained at higher levels compared to the controls, with only a slight reduction from 0.47 to 0.43 R.U. (Fig. 3a). Meanwhile, the humic-like fluorescence intensity in the control groups rose gradually from 0.14 ± 0.01 to 0.18 ± 0.01 R.U., whereas the values in the MC treatment groups were consistently $\sim 10\%$ higher than those in the control groups (Fig. 3b).

3.4. Microbial abundance and community structure

During the incubation period, microbial abundance in the control group exhibited a pronounced initial increase, rising from 1.57×10^9 to 3.91×10^9 cells/L over the initial 15 days. This growth subsequently decelerated and reached a stable equilibrium. In contrast, microbial proliferation in the MC treatment group was significantly slower, culminating in a final abundance of 1.97×10^9 cells/L. Consequently, the average microbial abundance in the MC treatment group was approximately 60% lower than that of the control group (Fig. 4a). High-throughput sequencing revealed a diverse bacterial community comprising 23 phyla, 43 classes, 99 orders, 143 families, and 193 genera. Analysis of the rank-abundance curve indicated a notable structural difference, with species richness (as indicated by rank) in the MC treatment group substantially exceeding that of the control group, with ranks of 380 and 150, respectively (Fig. S4a).

Furthermore, both PCA and ANOSIM revealed statistically significant differences ($p < 0.01$) in the microbial community composition

between the control and MC treatment groups (Fig. S4b). Within the control group, Alphaproteobacteria predominated as the major microbial class, with relative abundances ranging from 86% to 96%. Conversely, the MC treatment group demonstrated a pronounced shift in microbial dominance from Alphaproteobacteria to Gammaproteobacteria, which constituted 99% of the microbial community. Over time, the relative abundance of Gammaproteobacteria subsequently decreased, allowing for the succession of Bacteroidia, Bacilli, and Actinobacteria (Fig. 4b). The LEfSe further emphasized these differences through histograms of LDA scores, which highlighted significant variations in microbial taxa between the control and MC treatment groups across various taxonomic levels. Notably, LDA scores exceeding 4 were recorded for *c.Alphaproteobacteria*, *o.Rhodobacterales*, *f.Rhodobacteraceae*, *g.Dinoroseobacter*, and *g.Marivita* in the control group, as well as *c.Gammaproteobacteria*, *o.Burkholderiales*, *f.Comamonadaceae*, and *g.Delftia* in the MC treatment group. These taxa collectively contribute to the observed compositional divergences (Fig. 4c).

3.5. RDA results

This study examined the relationships between bacteria involved in carbon mineralization and key environmental variables. Notably, Alphaproteobacteria and Gammaproteobacteria, which exhibited significant class-level differences between control and MC treatment groups, demonstrated strong positive correlations with pH, DIC, DOC, Dsi concentration, and humic-like fluorescence intensity (Fig. 5a). These findings highlight the profound impact of environmental condition shifts on the succession of crucial microbial communities. To further elucidate these dynamics, RDA was employed to quantify the relationships and relative influences of physicochemical factors on microbial community composition. The results clearly indicate that MC facilitates microbial succession by modulating environmental parameters, particularly pH and DIC, during flocculation and algal removal processes, ultimately suppressing the rate of organic carbon mineralization (Fig. 5b).

4. Discussion

4.1. Enhanced carbon sequestration potential of blooming microalgae by MC

While blooming microalgae are acknowledged as potential carbon sinks (Sarwer et al., 2022; Sayre, 2010; Singh and Ahluwalia, 2013), a substantial portion of algal-derived organic carbon is subject to rapid

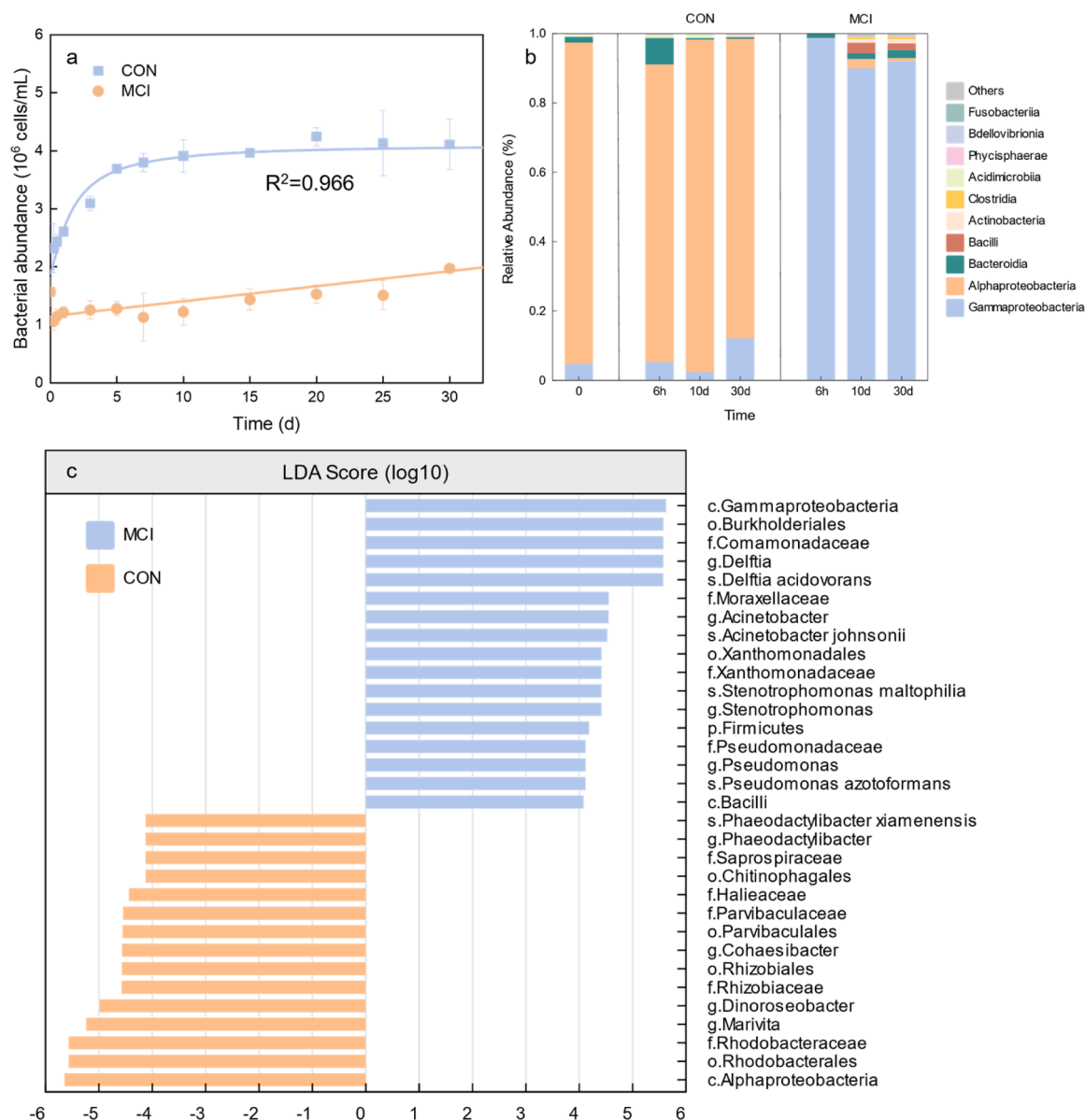


Fig. 4. Variations in microbial composition during batch incubation experiments with *S. costatum* (a: bacterial abundance; b: relative abundance of the top 10 bacterial species at the class level; c: LDA value distribution, where the LDA score represents the impact level of species with significant differences between the MCI and control groups).

microbial degradation due to its high bioavailability (Li et al., 2023; Raza et al., 2023; Williams et al., 2025). Presently, considerable uncertainties remain concerning the mineralization dynamics of algal-derived organic carbon and their implications for effective carbon sequestration (Yuan and Liu, 2021). In this study, following the decline of HABs, a substantial portion of the algal-derived organic carbon underwent rapid mineralization, thereby diminishing its sequestration potential. Notably, the addition of MC significantly inhibited the mineralization rate by more than 80% and reduced TOC mineralization by over 55%, as further evidenced by the observed decrease in DIC regeneration. These results provide empirical evidence that MC substantially enhances carbon sequestration potential. Similar to natural oceanic dust deposition, which enhances the BP efficiency (Kim et al., 2021), MC is anticipated to promote the vertical export of organic carbon while protecting organic matter through delayed degradation.

Recently, the mineral carbon pump has emerged as a promising strategy for enhancing oceanic carbon sequestration, garnering increasing attention (Xiao et al., 2025; Yuan and Liu, 2021).

Significantly, MC is widely recognized as an environmentally sustainable and the most extensively applied method for mitigating HABs in coastal waters (Anderson et al., 2025; Song et al., 2021; Yu et al., 2017). According to the China Marine Disaster Bulletin (2023), from 2014 to 2023, HABs covered an average area of 5,448 km², contributing an estimated 10^3 – 10^5 tons of organic carbon annually (Jiang et al., 2024). Over 50% of this algal-derived organic carbon is mineralized into DIC, which disperses into the water column and the atmosphere, resulting in an estimated annual carbon loss of up to 93,500 tons in coastal regions. This study suggests that the application of MC to mitigate HABs could potentially enhance annual marine carbon sequestration by up to 84,200 tons. However, such estimates of enhanced organic carbon burial are subject to uncertainties, as sediment resuspension is prevalent (Guo et al., 2024) and the potential remobilization of degraded organic carbon may lead to an overestimation of MC's contribution.

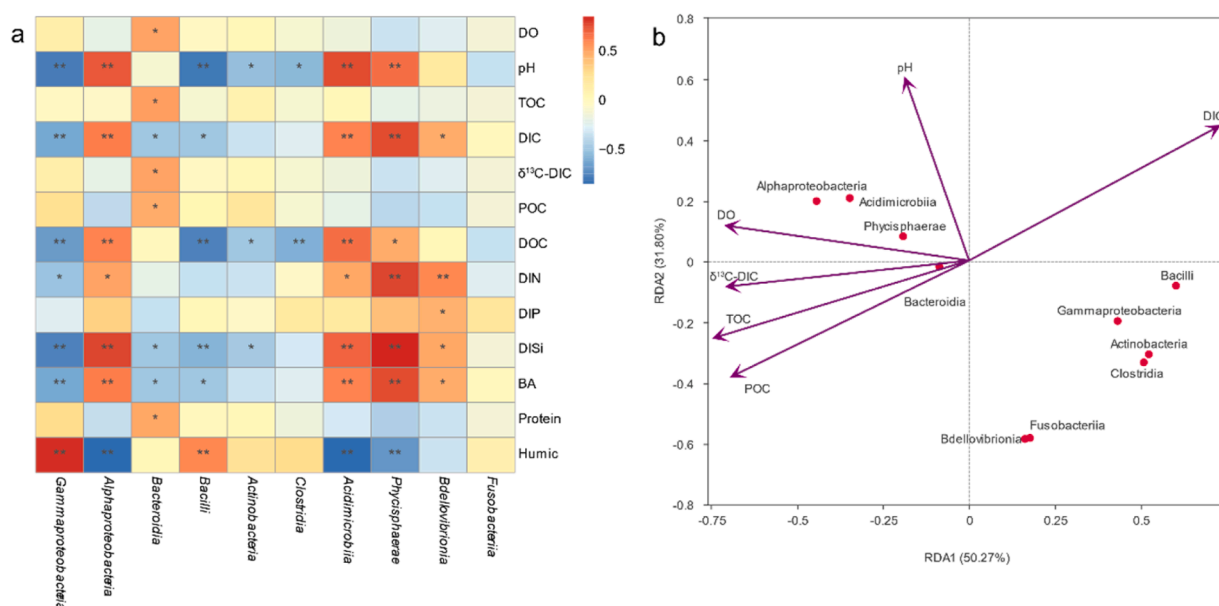


Fig. 5. Spearman correlation and RDA between environmental factors and bacterial community composition.

4.2. Coating algal cells with MC provide mineral protection and slows the mineralization process

The mineralization of marine organic carbon is a crucial process in regulating carbon cycling (Arndt et al., 2013). During the sedimentation of organic carbon produced by HABs, the majority of particulate organic carbon is rapidly mineralized by heterotrophic bacteria, with typically less than 30% ultimately sequestered in the underlying sediments (Burdige, 2007). Blattmann et al. demonstrated that the interaction between clay minerals and organic matter can significantly enhance the preservation and long-term burial of organic carbon in sediments. In this study, the application of MC substantially reduced the mineralization rate of algal-derived organic carbon, highlighting its strong potential to enhance carbon sequestration. A key mechanism involves the rapid flocculation of algal cells by MC, resulting in the formation of stable MC–algae flocs. Specifically, PAC–MC composites aggregated swiftly, entrapping algal cells and forming larger flocs (Fig. S6–S7). This process dramatically accelerated the sedimentation of algal-derived organic carbon and significantly enhanced BP efficiency. Consequently, the duration of organic carbon exposure to microbial degradation within the water column was significantly reduced, leading to a delay in mineralization. Furthermore, MC interacts organic carbon via physical adsorption and chemical interactions, promoting the polymerization of low-molecular-weight compounds into larger, more recalcitrant molecules (Xiao et al., 2025). These complexes are subsequently intercalated within clay interlayers, thereby enhancing their resistance to microbial degradation.

Furthermore, the coaggregation of algae and MC occurred readily, facilitating the formation of mineral-associated organic matter through the ‘direct sorption pathway’ (Sokol et al., 2019). In the control group, ruptured algal cells were uniformly distributed with a wrinkled morphology, remaining fully exposed (Fig. S8a). In contrast, within the MCI group, algal cells adhered tightly to the MC surface, forming small flocs or particle aggregates (Fig. S8b). These initial flocs were further interconnected by the long-chain polymeric structure of PAC, which facilitated the formation of larger aggregates through adsorption bridging, leading to the near-complete encapsulation of algal cells. Organic carbon can be effectively sequestered within clay micro-aggregates and interlayer spaces, as noted by Huang et al. 2020, thereby enhancing preservation and significantly improving vertical carbon export efficiency. Additionally, MC altered the biodegradability of

algal-derived organic matter, as indicated by the increased fluorescence intensity of humic-like substances. The stable binding between MC and organic carbon is mediated through hydrogen bonding (Feng et al., 2014; Gu et al., 2024), electrostatic interactions, ligand exchange (Hu et al., 2024), and cation bridging (Clarholm et al., 2015). These interactions are instrumental in the conversion of labile organic carbon into mineral-associated organic carbon (MAOC), subsequently decreasing its bioavailability and fostering the development of recalcitrant, inert organic matter. This process enhances the resistance of organic carbon to microbial mineralization (Kennedy et al., 2006; Lavallee et al., 2020)

4.3. MC delays the mineralization of alga-derived organic carbon by altering microbial communities

The processes of mineralization and sequestration of organic carbon are mediated by microbial activity (Feng et al., 2022; Raza et al., 2023), and are further modulated by variables including temperature (Heitkötter et al., 2017), dissolved oxygen and mineral availability (Xiao et al., 2025; Zhao et al., 2023). In this study, the logistic growth pattern observed in microbial abundance within the control group suggests that labile organic carbon fuels rapid microbial proliferation. This finding is consistent with prior reports of elevated heterotrophic bacterial densities in regions enriched with bioavailable organic matter (Guo et al., 2024). Moreover, protein-like substances are preferentially utilized by fast-growing heterotrophic microbes (Fontaine et al., 2003), as evidenced by the progressive decline in protein-like fluorescence intensity and the swift TOC mineralization in the control group. In contrast, the presence of MC significantly inhibited microbial growth and reproduction by forming MAOC, which reduces substrate bioavailability and exerts limited stimulation on bacterial activity.

More importantly, the compartmentalization of organic matter by MC can create distinct microsites (Kleber et al., 2021), as evidenced by alterations in the microbial community in this study. Specifically, there is a transition from Alphaproteobacteria, which dominated in the control group, to Gammaproteobacteria, known for their proficiency in metabolizing macromolecules (Zhang et al., 2018). LEfSe analysis further revealed pronounced compositional differences between the control and MC groups. The control group was enriched with chemo-organotrophic taxa dependent on easily degradable organic carbon, including *c. Alphaproteobacteria* and *o. Rhodobacterales* (Jiang et al., 2024;

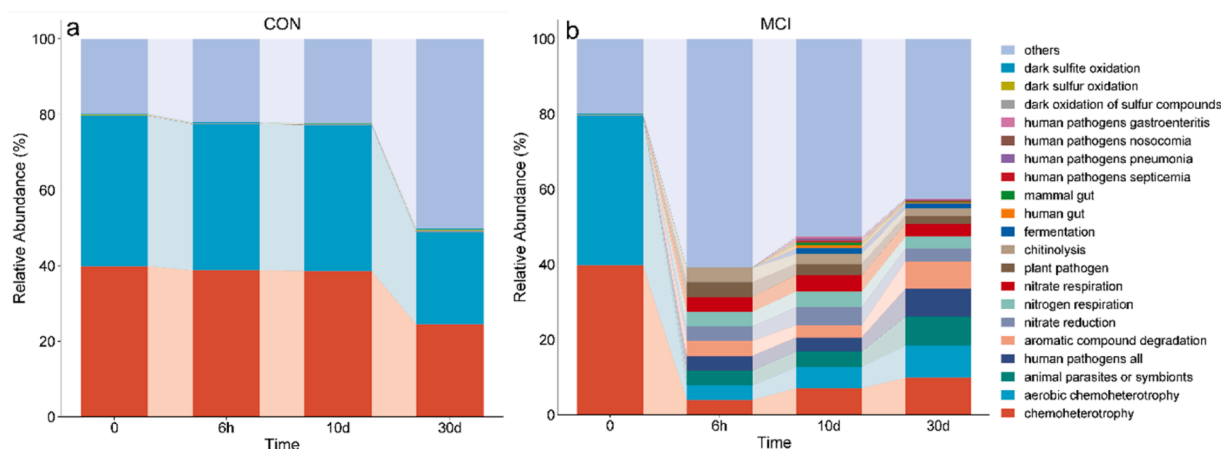


Fig. 6. Relative abundance of functional genes (a: control group; b: MCI group).

Simon et al., 2017; Wei et al., 2023). In contrast, the MC treatment group was characterized by taxa proficient in recalcitrant carbon degradation, such as *o.Burkholderiales* and *f.Comamonadaceae* (Braña et al., 2016; Das et al., 2014; Jiao, 2024). In this study, the FAPROTAX tool was used to predict microbial functions. The analysis demonstrated that algal-derived organic carbon was predominantly utilized by heterotrophic bacteria, with gene expression primarily associated with chemoheterotrophy and aerobic chemoheterotrophy (Fig. 6). These results suggest a highly active biomineralization process (Zhou et al., 2020). However, within the MC treatment group, the relative abundance of functional genes in these categories was significantly lower compared to the control group. Overall, the total relative abundance of mineralization-related functional genes, including those associated with chemoheterotrophy, aerobic chemoheterotrophy, and aromatic compound degradation, decreased by approximately 50% in the MC treatment group. Consequently, the application of MC suppressed the relative abundance of functional genes involved in microbial mineralization, thereby decelerating the mineralization rate of algal-derived organic carbon.

5. Conclusion and environment implications

This study provides direct evidence underscoring the inhibitory effects of MC on the mineralization and significant potential in enhancing carbon sequestration. Results from batch incubation experiments indicate that MC significantly retards the mineralization of algal-derived organic carbon. Specifically, MC rapidly adsorbs and flocculates algal cells, thereby enhancing the sedimentation of algal-derived organic carbon and effectively sequestering it. More importantly, the reduced bioavailability induced by MC impedes the growth and metabolic functions of bacteria responsible for carbon mineralization.

Under the combined influences of climate change and coastal eutrophication, there has been a significant global increase in both the scale and duration of HABs. This study introduces a method with practical applications for mitigating the ecological risks associated with HABs while simultaneously enhancing their carbon sequestration potential. Through the synergistic interactions among "organic carbon-MC-microorganisms," MC not only mitigates the detrimental impacts of HABs but also promotes the export of algae-derived organic carbon to the deep sea. This dual benefit of hazard mitigation and carbon sequestration is thus achieved. Furthermore, the economic feasibility of MC for large-scale application in HABs control appears promising, with estimated costs for MC in most scenarios being approximately 10,000 USD per km², excluding expenses related to vessel deployment and labor (Anderson et al., 2025; Yu et al., 2017). Further optimization of mineral ratios and components, along with modifications to clay properties such as surface charge, specific surface area, and hydrophobicity, holds

substantial potential for improving carbon sequestration efficiency (Kleber et al., 2021; Yuan and Liu, 2022). Additionally, further systematic investigation is required to evaluate the long-term eco-environmental effects of MC, including potential changes in seawater chemical properties and carbon cycling, in the field treatment of HABs in marine environment.

CRedit authorship contribution statement

Lianbao Chi: Writing – original draft, Investigation, Data curation. **Jing Chen:** Visualization, Investigation, Data curation. **Tianhao Zheng:** Software, Formal analysis. **Wentao Wang:** Software. **Xiuxian Song:** Supervision, Project administration, Funding acquisition, Conceptualization. **Zhiming Yu:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.wroa.2026.100493](https://doi.org/10.1016/j.wroa.2026.100493).

Data availability

Data will be made available on request.

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